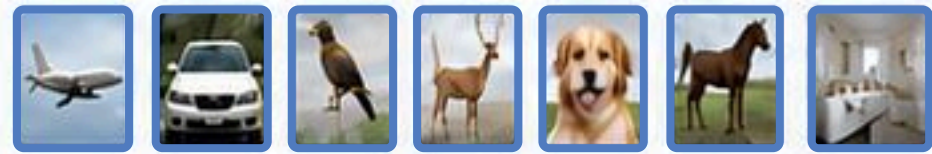


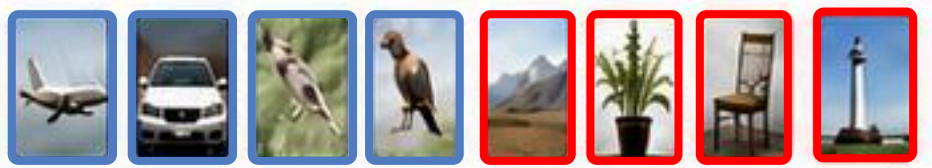
Inputs

$(x, y) \rightarrow \left(\text{img}, \text{"Bird"} \right)$

In-Distribution (InD) set $\mathcal{S}_{\text{in}}$ (labeled)



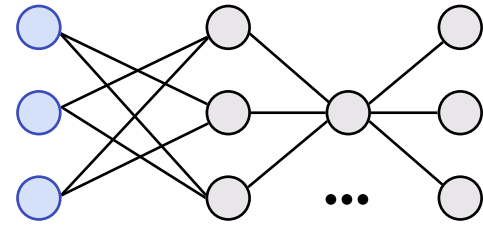
Unlabeled wild set $\mathcal{S}_{\text{wild}}$ (mixture)



■ In-Distribution (InD)
■ Out-of-Distribution (OOD)

Reference Gradient Estimation

Train InD classifier on $\mathcal{S}_{\text{in}}$



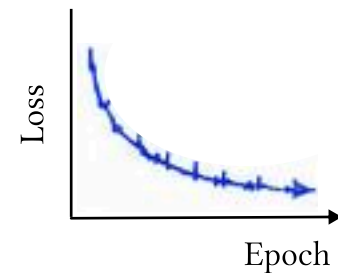
$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi} \frac{1}{n} \sum_{(x_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_{\phi}(x_i), y_i)$$

Mean InD reference gradient

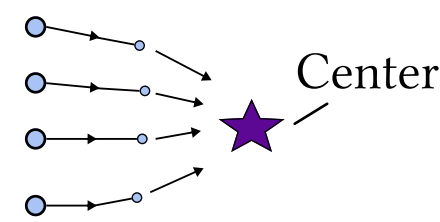
$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla_{\phi} \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(x_i), y_i)$$

Gradient Center

Train a classifier



Per-sample gradients (high-dimensional)



Why element-wise median (EWM)?

Affected by outliers vs. Robust to outliers

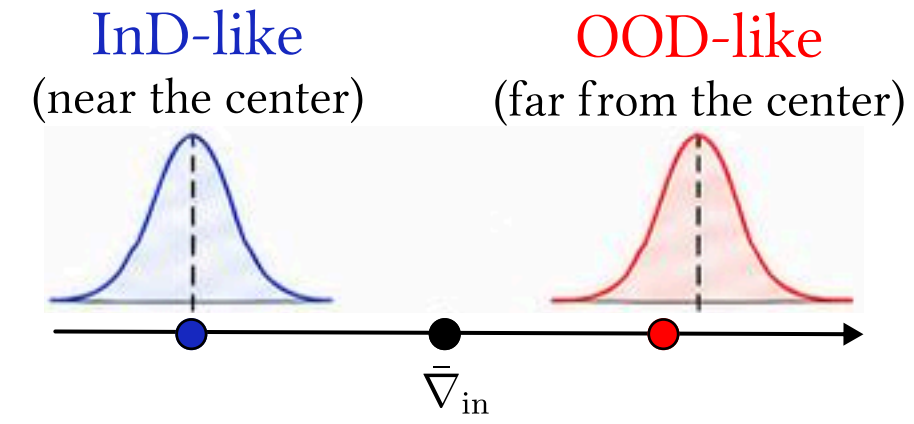
Mean vs. Median

Extract Candidate Outliers

Compute gradients for wild samples $\tilde{x}_i$

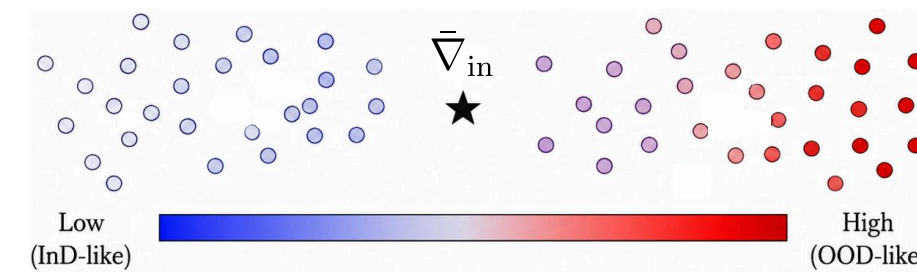
$$G_i = \nabla_{\phi} \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{x}_i), \hat{y}_{\tilde{x}_i})$$

$$\hat{y}_{\tilde{x}_i} = \arg \max_y f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{x}_i)_y$$



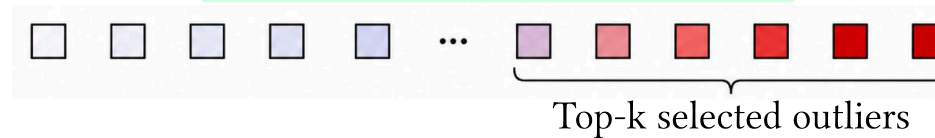
Distance-based Selection

EWM Gradients of wild samples $\text{EWM}(G_i)$



Rank by distance to $\bar{\nabla}_{\text{in}}$ and select top-k farthest

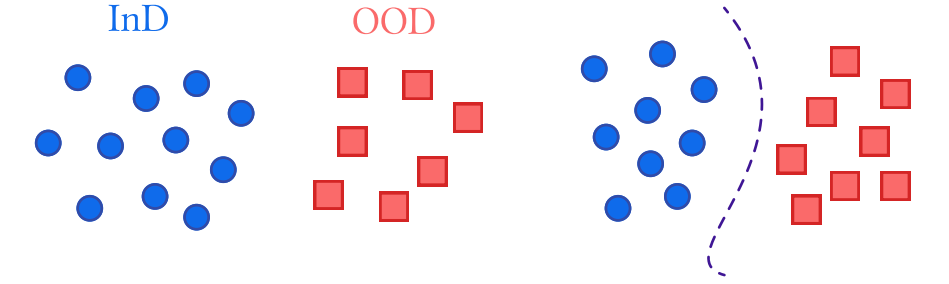
$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(\{G_i\}_{i \in \mathcal{S}_{\text{wild}}})\|_2$$



$$\hat{\mathcal{S}}_{\text{out}} = \text{Top-}k \text{ outliers}$$

OOD Detector

Training Data Learned Decision Boundary



$$g_{\theta} \leftarrow \text{Train}(\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}})$$

Test-time Inference

Input



OOD detector g_{θ}

→ OOD score

High score → Likely OOD
Low score → Likely InD